

Exact-gradient adaptation for AVC algorithm

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Abstract

The adaptive vibration cancellation (AVC) algorithm of Muradore, Pettazzi, Fedrigo & Clare (SPIE 8447-38) updates its state by a *pseudo*-gradient: it descends the squared prediction error while treating the regressor $\mathbf{W}$ as independent of the state, even though $\mathbf{W}$ depends on the state through the control parameters θ . We derive the exact gradient, including the dropped chain-rule term, in closed form. Two structural facts follow analytically and are confirmed numerically: (i) even the exact gradient leaves the plant-response parameters *unobservable* from a single tone (the update direction is confined to a two-dimensional per-cycle subspace, so the averaged information matrix has rank two); and (ii) applying the exact gradient to the *disturbance* parameters attenuates their update by a factor $\theta_{\min}/(|P|^2 + \theta_{\min})$, destroying the integral action that actually cancels the vibration. These motivate a *hybrid* update—exact gradient on the plant parameters, pseudo-gradient (integral action) on the disturbance parameters—which we implement and exercise in a closed-loop SPECULA single-conjugate AO simulation. In a controlled comparison (identical loop, identical atmosphere and vibration realisation, *only* the state update differing) the original pseudo-gradient with online plant adaptation delivers *no cancellation at all* (+0.1 dB on the 47 Hz tone relative to the no-AVC baseline) while its plant estimate wanders chaotically over three decades in magnitude and the full $\pm 180^\circ$ in phase, and its disturbance estimate grows without bound. The corrected hybrid, from the same seeds, cancels the tone by -48 to -59 dB and reduces the mode-0 residual RMS by 94–96%. Its plant estimate nonetheless makes only a bounded, *radial* (magnitude-only) adjustment—the phase never rotates toward the true value—an explicit picture of the unobservability, and the reason a measured, frozen plant remains the best-performing reference (-63 dB).

1 The algorithm

Following the paper, a single AVC instance tracks one sinusoidal disturbance at (angular) phase $\alpha(t)$. Its state collects the real and imaginary parts of the plant frequency response and of the disturbance quadratures,

$$\mathbf{x} = (x_1, x_2, x_3, x_4)^\top = (P_R, P_I, \lambda_c, \lambda_s)^\top. \quad (1)$$

From $\mathbf{x}$ it forms the control quadratures $\theta = (\theta_c, \theta_s)$ that would cancel the disturbance if $\mathbf{x}$ were correct (their eq. 11),

$$\theta_c = -\frac{x_1x_3 - x_2x_4}{D}, \quad \theta_s = -\frac{x_1x_4 + x_2x_3}{D}, \quad D = x_1^2 + x_2^2 + \epsilon, \quad (2)$$

where $\epsilon \geq 0$ is a regulariser on the otherwise singular denominator, discussed in the next paragraph. With $c \equiv \cos \alpha$, $s \equiv \sin \alpha$, the time-varying regressor (their eq. 9) is

$$\mathbf{W}(\alpha, \theta) = \begin{pmatrix} \theta_c c + \theta_s s \\ \theta_s c - \theta_c s \\ c \\ s \end{pmatrix}, \quad (3)$$

and the measurement model is $y = \mathbf{W}^\top \mathbf{x}$, an identity when $\mathbf{x}$ holds the true plant and disturbance. Writing the prediction (equation) error $e = \mathbf{W}^\top \hat{\mathbf{x}} - y$ and cost $J = \frac{1}{2}e^2$, the paper updates (their eq. 10, discretised with step $\mu = g_x T_s$)

$$\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} - \mu \mathbf{W} e. \quad (4)$$

The regulariser ϵ . The paper divides by $x_1^2 + x_2^2$ exactly, i.e. $\epsilon = 0$. That denominator is the squared magnitude of the *plant estimate*, so the control gain (2) diverges whenever the estimate approaches the origin—an event the update law does nothing to prevent. Implementations therefore guard it: ESO’s `updateAVC.m` freezes θ to $(1, 0)$ whenever $x_1^2 + x_2^2 < 10^{-2}$, a *discontinuous* clamp. We use instead the smooth equivalent $D = x_1^2 + x_2^2 + \epsilon$ with $\epsilon = \theta_{\min} = 10^{-2}$ by default (`theta_min_energy` and `soft_clamp` in the code), which bounds the gain by $\sim 1/\epsilon$ without introducing a switching boundary for the state to chatter against.

Although ϵ is a numerical safeguard absent from the published algorithm, it turns out to control the entire magnitude of the exact gradient: every component of $\mathbf{g}$ carries a factor ϵ (Secs. 3.2 and 3.3), and at $\epsilon = 0$ the exact gradient vanishes identically (Remark 3.3). It is therefore worth tracking explicitly rather than treating as an implementation detail.

2 Exact differentiation

Equation (4) is *not* the gradient of J . Because $\mathbf{W} = \mathbf{W}(\theta(\hat{\mathbf{x}}))$ depends on the estimate, the exact derivative of the error is

$$\frac{\partial e}{\partial x_i} = \frac{\partial(\mathbf{W}^\top \mathbf{x})}{\partial x_i} = W_i + \sum_j \frac{\partial W_j}{\partial x_i} x_j = (\mathbf{W} + J_{\mathbf{W}\mathbf{x}}^\top)_i, \quad (J_{\mathbf{W}})_{ji} = \frac{\partial W_j}{\partial x_i}, \quad (5)$$

(y is data, hence constant in $\hat{\mathbf{x}}$ at the current step). The paper keeps only W_i and drops the chain-rule term $J_{\mathbf{W}\mathbf{x}}^\top \mathbf{x}$. The *exact* update descends along

$$\mathbf{g} = \mathbf{W} + J_{\mathbf{W}\mathbf{x}}^\top \mathbf{x}. \quad (6)$$

The Jacobian. Only W_1, W_2 depend on $\mathbf{x}$ (through θ); $W_3 = c$, $W_4 = s$ are constant. Hence, using (3), $\partial W_1 / \partial x_i = c \partial_i \theta_c + s \partial_i \theta_s$ and $\partial W_2 / \partial x_i = c \partial_i \theta_s - s \partial_i \theta_c$, so the chain-rule term collapses to a two-term combination

$$(J_{\mathbf{W}\mathbf{x}}^\top)_i = x_1 \frac{\partial W_1}{\partial x_i} + x_2 \frac{\partial W_2}{\partial x_i} = A \partial_i \theta_c + B \partial_i \theta_s, \quad \begin{aligned} A &= x_1 c - x_2 s, \\ B &= x_1 s + x_2 c, \end{aligned} \quad (7)$$

with the derivatives of (2) (writing $\partial_i \equiv \partial / \partial x_i$)

$$\nabla \theta_c = \frac{1}{D} \begin{pmatrix} -x_3 - 2x_1 \theta_c \\ x_4 - 2x_2 \theta_c \\ -x_1 \\ x_2 \end{pmatrix}, \quad \nabla \theta_s = \frac{1}{D} \begin{pmatrix} -x_4 - 2x_1 \theta_s \\ -x_3 - 2x_2 \theta_s \\ -x_2 \\ -x_1 \end{pmatrix}. \quad (8)$$

Equations (6)–(8) give $\mathbf{g}$ analytically; they agree with a finite-difference Jacobian to $\sim 10^{-9}$ relative error, and the resulting SPECULA object reproduces an independent NumPy reference trajectory to $\sim 2 \times 10^{-17}$.

3 Two structural consequences

3.1 The plant stays unobservable (rank-2 information)

The question here is a concrete one: if the estimate $\hat{\mathbf{x}}$ is wrong, does the measurement notice? Directions in parameter space along which $\hat{\mathbf{x}}$ can be perturbed *without changing the predicted output at any phase* are invisible to any algorithm that only looks at that output, no matter how it descends. We show that a two-dimensional family of such directions always exists, and that it contains the plant.

Complex notation. The algebra becomes transparent in complex form. Collect the plant and the disturbance into two complex numbers, and the control into a third,

$$P = x_1 + ix_2, \quad \Lambda = x_3 + ix_4, \quad \vartheta = \theta_c + i\theta_s. \quad (9)$$

Since $(x_1 + ix_2)(x_3 + ix_4) = (x_1x_3 - x_2x_4) + i(x_1x_4 + x_2x_3)$, the two numerators of (2) are precisely the real and imaginary parts of $P\Lambda$, so (2) is the single statement

$$\vartheta = -\frac{P\Lambda}{D} \xrightarrow{\epsilon \rightarrow 0} -\frac{\Lambda}{\bar{P}}, \quad D = |P|^2 + \epsilon. \quad (10)$$

That is the design intent made explicit: the control is the disturbance divided by the plant, i.e. “whatever, once pushed through the plant, cancels Λ ”.

The measurement sees only one complex number. Writing the regressor (3) in the same notation and collecting the $\cos \alpha$ and $\sin \alpha$ parts of $\mathbf{W}^\top \mathbf{x}$ gives, after a one-line computation,

$$\boxed{\mathbf{W}^\top \mathbf{x} = \operatorname{Re}(Z) \cos \alpha + \operatorname{Im}(Z) \sin \alpha, \quad Z = \vartheta \bar{P} + \Lambda} \quad (11)$$

(verified numerically to machine precision). This is the crux. The model output depends on the four parameters (P, Λ) *only* through the single complex number Z — two real numbers. The map $(P, \Lambda) \mapsto Z$ goes from $\mathbb{R}^4$ to $\mathbb{R}^2$; for a given control ϑ it is linear in $(\bar{P}, \Lambda)$, so its kernel is exactly two-dimensional:

$$\delta Z = \vartheta \delta \bar{P} + \delta \Lambda = 0 \iff \delta \Lambda = -\vartheta \delta \bar{P}. \quad (12)$$

In words: *any* change in the plant estimate can be exactly compensated by a corresponding change in the disturbance estimate, leaving every measurement, at every phase, for all time, bit-for-bit unchanged. Choosing $\delta \bar{P} = 1$ and $\delta \bar{P} = i$ gives the two null directions explicitly,

$$\mathbf{n}_A = (1, 0, -\theta_c, -\theta_s)^\top, \quad \mathbf{n}_B = (0, 1, -\theta_s, +\theta_c)^\top, \quad (13)$$

each carrying unit weight on a *plant* axis. The ambiguity is physical, not numerical: with a single tone one observes one complex amplitude per cycle, while P and Λ together are two complex unknowns. The problem is underdetermined by exactly a factor two, and “how much does the plant scale my correction” cannot be separated from “how large is the disturbance”.

Why the exact gradient does not rescue this. Substituting (7) into (6) and grouping the c and s terms,

$$\begin{aligned} \mathbf{g} &= c \mathbf{v}_1 + s \mathbf{v}_2, \\ \mathbf{v}_1 &= \mathbf{u}_1 + x_1 \nabla \theta_c + x_2 \nabla \theta_s, \\ \mathbf{v}_2 &= \mathbf{u}_2 - x_2 \nabla \theta_c + x_1 \nabla \theta_s, \end{aligned} \quad (14)$$

with $\mathbf{u}_1 = (\theta_c, \theta_s, 1, 0)^\top$ and $\mathbf{u}_2 = (\theta_s, -\theta_c, 0, 1)^\top$ (so that $\mathbf{W} = c \mathbf{u}_1 + s \mathbf{u}_2$). The vectors $\mathbf{v}_1, \mathbf{v}_2$ depend on the state but *not* on the phase α . Hence, if the estimate is quasi-static over one

disturbance cycle (true for small step sizes, the regime of interest), $\mathbf{g}$ sweeps only the plane $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ and, averaging with $\langle c^2 \rangle = \langle s^2 \rangle = \frac{1}{2}$, $\langle cs \rangle = 0$,

$$\langle \mathbf{g} \mathbf{g}^\top \rangle = \frac{1}{2} (\mathbf{v}_1 \mathbf{v}_1^\top + \mathbf{v}_2 \mathbf{v}_2^\top), \quad \text{rank} \leq 2. \quad (15)$$

The information matrix is rank two whichever update direction is used: correcting the gradient changes *which* plane is explored, not the fact that only a plane is explored. Numerically, with ϑ held fixed the two remaining eigenvalues are zero to machine precision ($\sim 10^{-14}$ against $\mathcal{O}(1)$).

A caveat, and why excitation does not help either. Equation (15) is exact for constant ϑ . If ϑ varies in time the kernel (12) rotates with it, and comparing epochs with different ϑ *would* in principle determine P : from $Z^{(1)} - Z^{(2)} = (\vartheta^{(1)} - \vartheta^{(2)}) \bar{P}$ at constant Λ one could solve for $\bar{P}$. This is exactly what the offline probe measurement of the plant does, and it is why that procedure works. It does *not* rescue the online algorithm, for two reasons: the instantaneous gradient carries no memory across epochs, so it cannot form such differences; and ϑ is not an independent probe but is slaved to the estimate through (10), so it stops varying precisely as the algorithm converges. Deliberately modulating the *disturbance* (amplitude or phase) leaves the two extra eigenvalues at $\sim 10^{-5}$ relative — non-zero, but five orders below the observable pair and useless on any practical timescale. Observability must come from the command side, not the disturbance side.

3.2 The exact gradient annihilates the disturbance drive

For the disturbance components the chain-rule term nearly cancels the direct term. Using $\partial \theta_c / \partial x_3 = -x_1/D$, $\partial \theta_s / \partial x_3 = -x_2/D$ in (7),

$$(J_{\mathbf{W}\mathbf{x}}^\top)_3 = -\frac{Ax_1 + Bx_2}{D} = -\frac{(x_1^2 + x_2^2)}{D} c, \quad (16)$$

so that

$$\boxed{g_3 = c - \frac{x_1^2 + x_2^2}{D} c = c \frac{\epsilon}{D} = \cos \alpha \frac{\theta_{\min}}{|P|^2 + \theta_{\min}}}, \quad g_4 = \sin \alpha \frac{\theta_{\min}}{|P|^2 + \theta_{\min}}. \quad (17)$$

(Verified to machine precision.) The exact-gradient disturbance drive is the pseudo-gradient $(\cos \alpha, \sin \alpha)$ *attenuated by* $\theta_{\min}/|P|^2$: with $\theta_{\min} = 10^{-2}$ and $|P|^2 \sim 52$ in the demo this is $\sim 2 \times 10^{-4}$, i.e. the disturbance update is effectively switched off.

The reason is structural, not numerical. The control $\theta = -\lambda_{\text{est}}/P$ is *designed* so that raising the disturbance estimate raises the cancelling control by the same amount; increasing λ_{est} feeds a direct term ($+c \delta \lambda$ in $\mathbf{W}^\top \mathbf{x}$) and an indirect term through θ ($-c \delta \lambda$) that almost annihilate. The chain-rule term is exactly this indirect path. But minimising the equation error is the *wrong* objective for the disturbance: what we want is to drive the *physical residual* y to zero, and there the pseudo-gradient is exactly right. With the plant fixed, (4) restricted to x_3, x_4 reads

$$x_3 \leftarrow x_3 - \mu \cos \alpha e \implies \frac{d\lambda_{\text{est},c}}{dt} \propto -(\lambda_{\text{est},c} - \lambda_c), \quad (18)$$

i.e. a stable *integral controller* that drives $\lambda_{\text{est}} \rightarrow \lambda$ and hence $y \rightarrow 0$. The disturbance subproblem is a well-posed linear regression against the persistently exciting, observable regressor $(\cos \alpha, \sin \alpha)$; the plain correlation is the correct estimator, and the chain-rule term must *not* be added.

3.3 The exact plant update is radial, and of order ϵ

The same collapse happens on the plant side, in a more striking way. Carrying the complex notation of Sec. 3.1 through (7)–(8) (using $A + iB = Pe^{i\alpha}$, $\vartheta = -P\Lambda/D$, and $\Lambda e^{-i\alpha} + \bar{\Lambda}e^{i\alpha} = 2\text{Re}(\bar{\Lambda}e^{i\alpha})$) the two plant components of the exact gradient collapse to the closed form

$$g_1 + i g_2 = -2 \mathcal{R} \frac{\epsilon}{D^2} P, \quad \mathcal{R} \equiv \text{Re}(\bar{\Lambda} e^{i\alpha}) \in \mathbb{R} \quad (19)$$

(verified to machine precision for several ϵ). Two consequences, both directly visible in the closed-loop runs of Sec. 5:

- **The update is exactly radial.** $g_1 + i g_2$ is a *real* multiple of P : the exact plant update can only lengthen or shorten the plant estimate, never rotate it. The phase $\arg P$ is a conserved quantity of the exact plant update. Numerically the cross product $g_1 x_2 - g_2 x_1$ vanishes to $\sim 10^{-13}$ relative, whereas for the pseudo-gradient the same quantity is of order unity—which is why the original algorithm’s plant phase wanders over the full $\pm 180^\circ$ (Fig. 1c) while the hybrid’s sits exactly at its seed (Fig. 2). What looked empirically like “the update cannot rotate $\arg P$ ” is a theorem.
- **The update is $\mathcal{O}(\epsilon)$.** It is proportional to the regulariser $\epsilon = \theta_{\min}$ of (2). With $\epsilon = 0$ — the paper’s formula — the exact gradient does not move the plant estimate *at all*.

Why both components are $\mathcal{O}(\epsilon)$: the prediction is $\mathcal{O}(\epsilon)$. Equations (17) and (19) share the factor ϵ for one reason. Substituting the slaved control (10) into (11) with $\mathbf{x} = \hat{\mathbf{x}}$,

$$\hat{Z} = \vartheta \bar{P} + \hat{\Lambda} = -\frac{\hat{P}\hat{\Lambda}}{D} \bar{P} + \hat{\Lambda} = \hat{\Lambda} \left(1 - \frac{|\hat{P}|^2}{D}\right) = \hat{\Lambda} \frac{\epsilon}{D} \xrightarrow{\epsilon \rightarrow 0} 0. \quad (20)$$

The predicted output is identically zero when $\epsilon = 0$: by construction, the control ϑ is the one that cancels $\hat{\Lambda}$, so the model predicts perfect cancellation of its own estimate. Consequently the prediction error is simply $e = \mathbf{W}^\top \hat{\mathbf{x}} - y \simeq -y$, the negated physical residual, and the exact gradient—the derivative of a quantity that is identically zero—is itself $\mathcal{O}(\epsilon)$.

Remark (the $\epsilon \rightarrow 0$ collapse). Taking the limit in (17) and (19) simultaneously,

$$\epsilon = 0 \implies \mathbf{W}^\top \hat{\mathbf{x}} \equiv 0 \text{ for every } \hat{\mathbf{x}} \implies J = \frac{1}{2} (\mathbf{W}^\top \hat{\mathbf{x}} - y)^2 = \frac{1}{2} y^2 \implies \mathbf{g} \equiv \mathbf{0}. \quad (21)$$

With the paper’s exact formula, the cost being minimised *does not depend on the parameters being optimised at all*: θ is constructed to cancel $\hat{\Lambda}$, so the model always predicts perfect cancellation of its own estimate, whatever that estimate is. A cost independent of its arguments has zero gradient, and “exact gradient descent” on it is a no-op. Numerically, at $\epsilon = 0$ we measure $\|\mathbf{g}\| \sim 4 \times 10^{-15}$ (machine zero) while $\|\mathbf{W}\| \sim 20\text{--}40$.

This is the sharpest statement of the paper’s “error”: the exact gradient of the equation error is not merely a poor update, it is (up to the regulariser) *no update*. The pseudo-gradient is therefore not an approximation to the exact gradient but a different, useful object—the correlation of the physical residual with the regressor, i.e. LMS/integral action. It also explains, retrospectively, the **full exact** row of Table 3: with both halves receiving $\mathcal{O}(\epsilon)$ drives, the algorithm barely adapts at all, the disturbance estimate never leaves its initial value, and no cancellation occurs. By the same token, the modest plant adaptation that the hybrid *does* perform exists only by virtue of $\epsilon \neq 0$: it is an artefact of the safeguard, not of the algorithm as published.

3.4 The hybrid update

Sections 3.1–3.3 motivate matching each half of the state to its correct objective:

$$\hat{x}_{1,2} \leftarrow \hat{x}_{1,2} - \mu c_p g_{1,2} e \quad (\text{exact}), \quad \hat{x}_{3,4} \leftarrow \hat{x}_{3,4} - \mu W_{3,4} e \quad (\text{pseudo / integral}), \quad (22)$$

implemented as `exact_gradient: plant` in the SPECULA AVC object (c_p is the plant-update damping constant c). This gives the plant a true, non-diverging descent direction while leaving the disturbance integrator intact.

3.5 Why the plant update needs damping (c_p)

The plant components need their own, much smaller step size. There are two distinct reasons, one for each update variant, and it is worth separating them because they behave differently.

(i) Pseudo-gradient: the loop gain grows as the canceller works. For a gradient step $\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} - \mu M \mathbf{g} e$ with $M = \text{diag}(c_p, c_p, 1, 1)$, linearising about a stationary point gives $\delta \mathbf{x} \leftarrow (I - \mu M \langle \mathbf{g} \mathbf{g}^\top \rangle) \delta \mathbf{x}$, so stability requires the eigenvalues of $\mu M \langle \mathbf{g} \mathbf{g}^\top \rangle$ to lie in $(0, 2)$. For the pseudo-gradient this is computable in closed form: from $\mathbf{W} = c \mathbf{u}_1 + s \mathbf{u}_2$ with $\mathbf{u}_1 \cdot \mathbf{u}_2 = 0$ and $|\mathbf{u}_1|^2 = |\mathbf{u}_2|^2 = |\vartheta|^2 + 1$,

$$\langle \mathbf{W} \mathbf{W}^\top \rangle = \frac{1}{2} (|\vartheta|^2 + 1) \Pi, \quad \lambda_{\max} = \frac{1}{2} (|\vartheta|^2 + 1), \quad (23)$$

with Π the projector onto $\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$ (verified numerically: the spectrum is $\{\lambda_{\max}, \lambda_{\max}, 0, 0\}$ to machine precision). Because $|\vartheta| \gg 1$ in practice, that subspace lies almost entirely along the plant axes, so the binding constraint is the plant one:

$$\mu c_p \frac{1}{2} (|\vartheta|^2 + 1) < 2 \quad \Longleftrightarrow \quad c_p < \frac{4}{g_x T_s (|\vartheta|^2 + 1)}. \quad (24)$$

The essential point is *not* the constant but the scaling. By (10), $|\vartheta| = |\hat{\Lambda}|/|\hat{P}|$ grows from 0 at start-up (no correction yet) to a large value once the canceller has built up its correction — in the demo $|\vartheta| \approx 82$, i.e. $|\vartheta|^2 + 1 \approx 6.9 \times 10^3$. **The plant-loop gain therefore grows by orders of magnitude as cancellation improves:** a c_p that is perfectly safe during the first transient can violate (24) later, when the algorithm is working well. This is a genuine trap, and it is one plausible reading of why the published algorithm can appear stable in short tests and diverge in long ones. With the demo numbers ($g_x = 3$, $T_s = 1 \text{ ms}$), (24) gives $c_p < 0.19$.

(ii) Exact/hybrid gradient: a state-dependent blow-up near $|P| \rightarrow 0$. For the hybrid the plant drive is (19) rather than $\mathbf{W}$, and its magnitude is

$$|g_1 + i g_2| = 2|\mathcal{R}| \frac{\epsilon |P|}{D^2}, \quad D = |P|^2 + \epsilon. \quad (25)$$

This is small when $|P|^2 \gg \epsilon$ (the $\mathcal{O}(\epsilon)$ regime of Sec. 3.3) but *grows* as $|P|$ shrinks, peaking near $|P|^2 \sim \epsilon$ where it behaves like $2|\mathcal{R}|/(\epsilon^{1/2})$ — larger by $\sim 1/\epsilon$ than in the well-conditioned regime. Since the update is radial, a shrinking $|P|$ increases its own update gain: the mechanism is a state-dependent runaway rather than a fixed-gain instability, and c_p is what keeps the trajectory out of it.

Practical determination. Equation (24) is derived for a delay-free, noise-free loop and should be treated as an order-of-magnitude guide, not a threshold: the real loop adds three frames of delay and measurement noise, both of which tighten it. The gap is easily seen — in the idealised (delay-free) model the hybrid remains bounded even at $c_p = 1$, whereas in the closed loop $c_p = 1$ diverges. In practice c_p is set empirically; the sweep below (closed loop, $g_x = 3$) shows a clear threshold between 0.1 and 1, consistent with the 0.19 predicted by (24).

c_p	plant estimate $ x_1+ix_2 $	47 Hz residual
1.0	diverges ($\rightarrow 309$)	470 nm (no cancellation)
0.1	7.22 $\rightarrow$ 6.36, settles	2 nm
0.01	7.22 $\rightarrow$ 7.13	11 nm
0.001	essentially frozen	12 nm

Table 1: Empirical determination of the plant-update damping in the closed loop (hybrid update, $g_x = 3$). Too large diverges; too small reduces to the frozen-plant case. Predicted bound (24): $c_p < 0.19$.

4 Idealised experiments

These use only the algorithm’s equations (no AO loop, no delay, no noise, known fixed frequency, measurement $y = \mathbf{W}^\top \mathbf{x}_\star$), so any pathology is intrinsic to the algorithm.

Observability (`avc_observability_check.py`). With θ frozen at the truth the update is linear, and the averaged information matrix has eigenvalues (1.777, 1.777, 1.8×10^{-14} , 1.8×10^{-14}): two non-zero, two zero to machine precision (rank 2). The observable component of the estimate error is driven to $\sim 10^{-13}$ while the unobservable component is frozen at its initial value. Seeding the plant at the true value and running the full (non-linear) update, the estimate nonetheless diverges (magnitude $0.55 \rightarrow$ thousands): unobservable drift plus the $1/(x_1^2 + x_2^2)$ feedback. Freezing the plant and adapting only x_3, x_4 recovers the true disturbance exactly ((1000.000, -0.000) vs. (1000, 0); error $\sim 2 \times 10^{-6}$).

Pseudo vs. exact (`avc_chainrule_check.py`). Table 2 contrasts the two update directions from a slightly wrong plant seed. The pseudo-gradient *actively wanders*; the exact gradient *settles* (a true gradient vanishes on the zero-error manifold) but at a wrong value—neither identifies the plant, consistent with Sec. 3.1.

update	final plant error	$\text{eig}(\mathbf{g}\mathbf{g}^\top)$	plant identified?
pseudo (chain term dropped)	0.265 (drifts up from 0.071)	[1.73, 1.73, $\sim 0, \sim 0$]	no (wanders)
exact (chain term kept)	0.063 (settles)	[$\sim 0, \sim 0, \sim 0, \sim 0$]	no (wrong value)

Table 2: Idealised single-tone adaptation; small amplitude for a stable step size.

5 Closed-loop SPECULA demonstration

The full demo is a single-conjugate AO loop (64 px pupil, 8×8 Shack–Hartmann, 40 Zernike modes, 1 kHz) with a standard integrator (gain 0.5, 2-frame delay) on all modes, a first-order 80 Hz actuator low-pass, and a 47 Hz tip vibration. AVC runs in parallel on mode 0, its command summed with the integrator’s.

Where AVC is useful. The integrator rejects low frequencies but, above its bandwidth, the sensitivity $|S| = |1/(1 + H_{OL})|$ exceeds unity (waterbed). The measured/analytic rejection is $-38, -24, -17, -9$ dB at 1, 5, 10, 20 Hz but strongly positive at 47 Hz: the injected 120 nm tone emerges as a 891 nm residual, an amplification of $7.4\times$ (+17.4 dB). This is precisely where a narrow-band canceller earns its place—the main controller actively makes the disturbance worse and cannot be retuned to fix it without sacrificing low-frequency performance.

5.1 Controlled comparison: original vs. corrected update

To isolate the effect of the gradient alone, six 10 s runs were performed that are *identical in every respect*—same seeded atmosphere, same vibration realisation, same loop, same $g_x = 3$, $g_\omega = 0.2$, $c_p = 0.1$, same 45 Hz initial frequency guess—differing *only* in the state-update rule and the plant seed. Two seeds are used: the measured closed-loop value ($|P| = 7.22 \angle 169.5^\circ$, “correct”) and a deliberately wrong one ($|P| = 1.41 \angle 135^\circ$). Results are collected in Table 3 and Fig. 1; steady-state statistics are taken over the second half of each run.

update	plant seed	mode-0 RMS	all-mode RMS	47 Hz amp.	vs. base	final $ x_1+ix_2 $
none (baseline)	—	630.4 nm	99.9 nm	891 nm	—	—
ORIGINAL pseudo	wrong	640.7 nm	101.5 nm	898 nm	+0.1 dB	205 (wandering)
ORIGINAL pseudo	correct	639.6 nm	101.3 nm	896 nm	+0.0 dB	64 (wandering)
CORRECTED hybrid	wrong	36.5 nm	7.3 nm	3.5 nm	−48.2 dB	13.8 (settled)
CORRECTED hybrid	correct	27.5 nm	6.2 nm	1.0 nm	−58.9 dB	4.0 (settled)
frozen plant	correct	26.5 nm	6.1 nm	0.6 nm	−63.0 dB	7.22 (fixed)

Table 3: Controlled comparison; only the state update and plant seed differ. The original pseudo-gradient with online plant adaptation provides no cancellation whatsoever (it is marginally *worse* than doing nothing); the corrected hybrid cancels by 48–59 dB. A measured, frozen plant remains the best performer.

Three observations follow.

- **The original update fails outright when the plant is adapted online**, and the failure is insensitive to the seed: starting from the exactly measured plant is no better than starting from a deliberately wrong one (+0.0 vs. +0.1 dB). This is the signature of a divergence driven by the update law itself, not of a poor initialisation. Panel (e) shows it is in fact mildly harmful: the 47 Hz peak is *larger* than the baseline and the broadband floor is raised, because a large, wrongly-phased correction is being injected.
- **The corrected hybrid works from either seed**, differing only in convergence time and final depth (−48.2 dB from the wrong seed, −58.9 dB from the correct one; cf. panel (a), where the wrong-seed trace takes ~ 5 s rather than ~ 2 s to converge). Its disturbance estimate plateaus rather than diverging.
- **Neither variant identifies the plant.** The hybrid’s estimate settles at $|P| = 4.0$ (correct seed) or 13.8 (wrong seed), neither being the true 7.22, and panel (c) shows the phase pinned at whatever value it was seeded with. Cancellation is nevertheless excellent, because the disturbance integrator compensates for any stable, correctly-*phased* plant estimate. The residual gap to the frozen reference (−58.9 vs. −63.0 dB) is the price of letting the magnitude drift at all.

5.2 Convergence timescales from a wrong seed

The wrong-seed hybrid run of Table 3 separates three distinct timescales, shown in Fig. 2:

- **Plant magnitude** rescales $1.41 \rightarrow 13.8$ within a fraction of a second, then settles—*not* at the true 7.22.
- **Plant phase** is frozen at the seed 135.000° (to three decimals, over 10 s): the hybrid update is effectively *radial*, adjusting $|P|$ but never $\arg P$. This is the rank deficiency of Sec. 3.1 made directly visible.

Original (pseudo-gradient) vs corrected (exact-gradient hybrid) AVC
 identical closed loop, 120 nm @ 47 Hz vibration, only the state-update differs

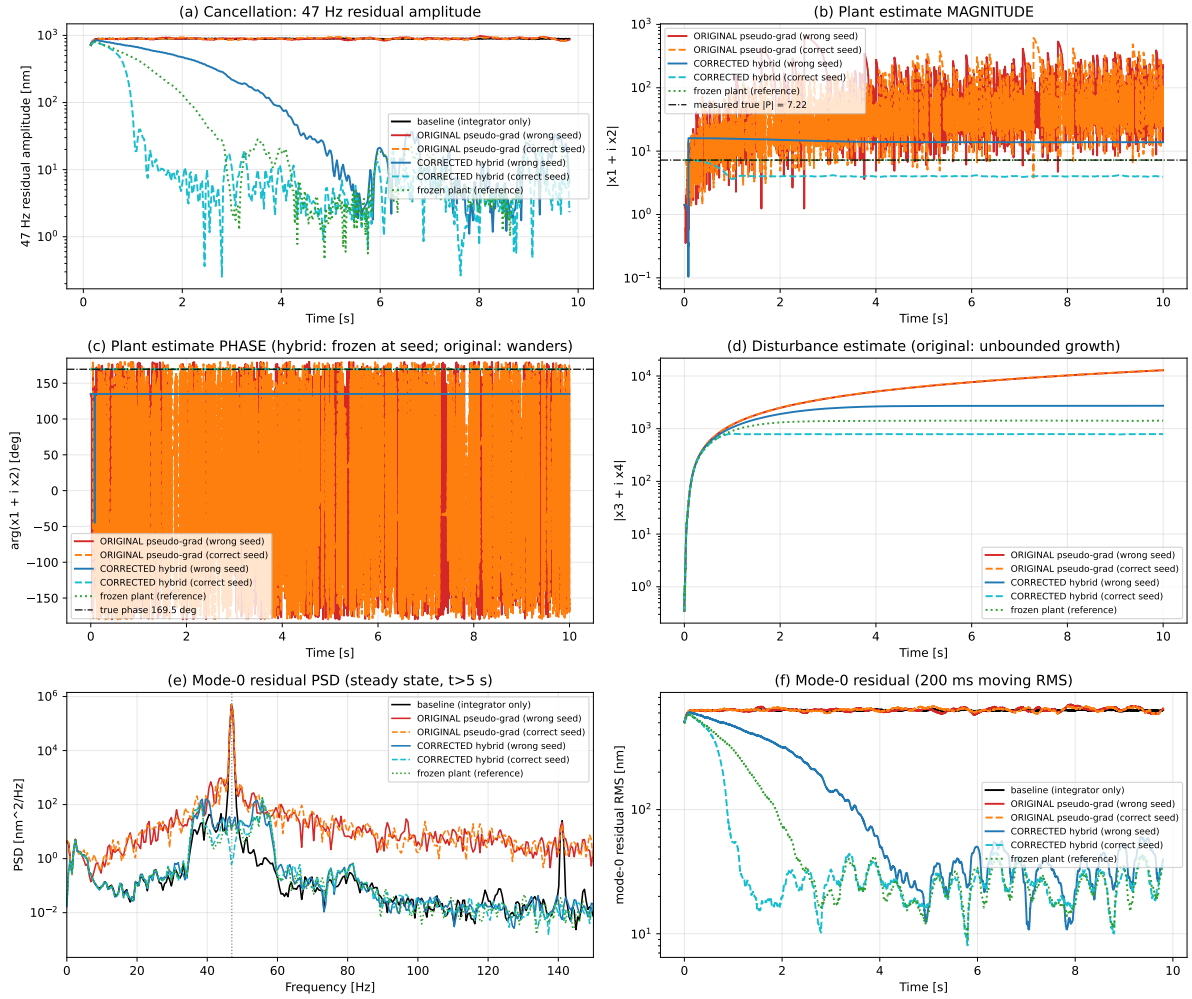


Figure 1: Original vs. corrected AVC under identical conditions. (a) the two ORIGINAL traces sit on the baseline—no cancellation—while the corrected hybrid and the frozen reference fall by 2–3 decades; (b),(c) the original plant estimate wanders over three decades in magnitude and fills the full $\pm 180^\circ$ in phase, whereas the hybrid settles and holds its seed phase exactly; (d) the original disturbance estimate grows without bound ($\rightarrow 1.3 \times 10^4$ and still rising at 10 s) while the hybrid plateaus; (e) in the steady-state PSD the original *enlarges* the 47 Hz peak and raises the broadband floor; (f) mode-0 residual.

- **Cancellation** nonetheless converges, on the disturbance-loop timescale set by g_x (~ 3.9 s to a $10\times$ reduction): the 47 Hz residual falls $891 \rightarrow 6.8$ nm, because the integral loop on x_3, x_4 compensates for whatever stable, correctly-phased plant estimate it is handed.

The frequency loop is a fourth, independent timescale: it locks from the wrong 45 Hz guess to 46.98 Hz with 0.02 Hz jitter, driven by the quadrature error $\sin \alpha e$ and requiring no plant knowledge at all—which is why frequency tracking succeeded throughout this study even when plant estimation did not.

6 Discussion

The two halves of the AVC state require different objectives, and the hybrid assigns each its correct one. The *disturbance* parameters form an observable linear regression whose goal is to null the physical residual; the pseudo-gradient is a certainty-equivalence integral controller that does exactly this, and the exact gradient sabotages it via (17). The *plant* parameters are the nonlinear, self-referential, single-tone-unobservable part where the pseudo-gradient’s neglect of $\partial \mathbf{W} / \partial \mathbf{x}$ causes active divergence; the exact gradient supplies a genuine descent direction that merely settles. Equivalently, the hybrid is a filtered-X-LMS structure: an LMS/integral canceller whose reference is filtered by the plant estimate (through θ), with the plant adapted on a separate, damped, stabilised path and never fed the term that would break the canceller.

Three practical corollaries follow. First, **the published update law must not be run with online plant adaptation**: Table 3 shows it then yields no cancellation at all, and does so regardless of how well the plant is initialised—the failure is in the update, not the initialisation. The paper’s own practice (seeding with approximate closed-loop plant values and relying on fast error nulling) is much closer to our “frozen” configuration, which is also the best performer here; read that way, our results are consistent with the paper’s reported success rather than contradicting it.

Second, **the plant is not identifiable online from a single tone**. Seed its *phase* approximately correctly—from a calibration, a probe measurement, or the analytic model—because no variant of the update can rotate it; the *magnitude* is forgiving, since the disturbance integrator absorbs a scale error.

Third, for a static plant the simplest robust choice remains a frozen, measured plant (-63.0 dB here). The corrected hybrid costs ~ 4 dB relative to that but adds the ability to track slow plant changes without the runaway; it is the appropriate choice when the plant drifts (thermal or elevation-dependent effects over a night) and recalibration is impractical.

Reproducibility. Derivation checks: `avc_chainrule_check.py` (finite-difference and reference-trajectory validation of the Jacobian and of Eq. (17)), `avc_observability_check.py` (rank-2 information matrix, drift). Closed-loop configurations: `params_scao_avc_full_demo.yml` (AVC; the update variant is selected by the `adapt_plant` and `exact_gradient` keys of the `avc:` block), `params_scao_vibration_baseline.yml` (integrator only), `params_measure_plant.yml` (closed-loop plant measurement by probe injection and lock-in). The six runs of Table 3 share the atmospheric seed (`AtmoEvolution seed=1`), so they differ only in the AVC update; per-run statistics and Figs. 1–2 are produced by the comparison scripts described in `README.md`, and the standard five-panel diagnostic by `analyze_results.py`.

Software. The update variants are implemented in `specula/processing_objects/avc.py`: `adapt_plant` (freeze/adapt x_1, x_2), `exact_gradient` $\in \{\text{false}, \text{'plant'}, \text{'full'}\}$ (which components receive the chain-rule term), and `soft_clamp` (the ϵ regulariser of Eq. (2)). The defaults (`exact_gradient: false`, `adapt_plant: true`, `soft_clamp: false`) reproduce the published algorithm bit-for-bit and are covered by a Matlab-reference regression test.

Corrected hybrid AVC: convergence timescales from a WRONG plant seed
seed $|P|=1.41$ @135deg ; measured true plant $|P|=7.22$ @169.5deg

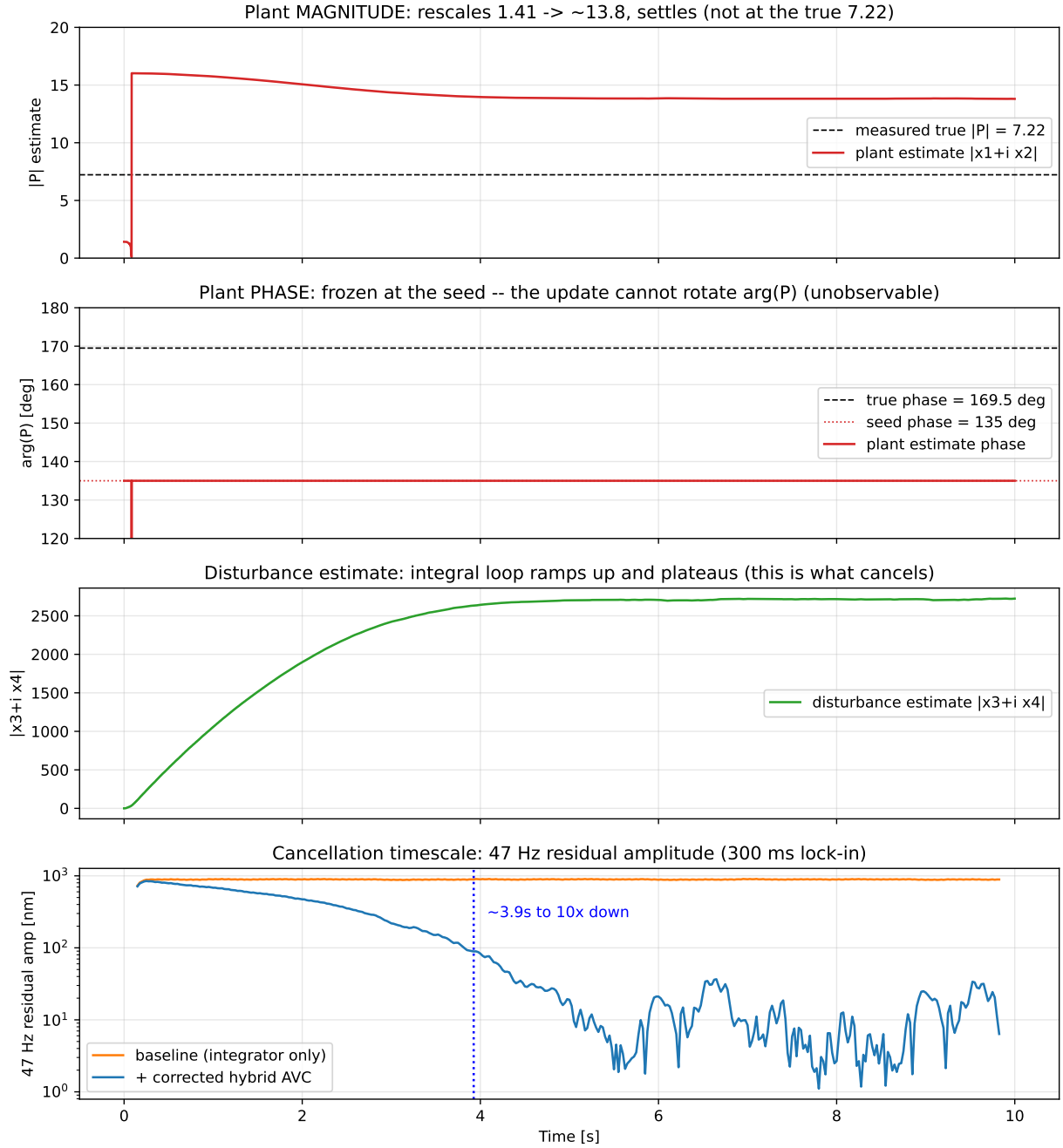


Figure 2: Corrected hybrid AVC from a wrong plant seed. Top to bottom: plant magnitude (rescales, misses the true value); plant phase (frozen at the seed—no rotation, unobservable); disturbance estimate (the integral ramp that does the cancelling); 47 Hz residual amplitude, baseline vs. AVC, ~ 3.9 s to 10 $\times$ down.

A Step-by-step computation of the Jacobian

This appendix expands Sec. 2 in full so the algebra can be checked line by line. Every entry below was verified against a finite-difference Jacobian (agreement $\sim 10^{-8}$).

Notation. With $c \equiv \cos \alpha$, $s \equiv \sin \alpha$ (constants in $\mathbf{x}$),

$$D = x_1^2 + x_2^2 + \epsilon, \quad N_c = -(x_1 x_3 - x_2 x_4), \quad N_s = -(x_1 x_4 + x_2 x_3), \quad \theta_c = \frac{N_c}{D}, \quad \theta_s = \frac{N_s}{D}, \quad (26)$$

and $W_1 = \theta_c c + \theta_s s$, $W_2 = \theta_s c - \theta_c s$, $W_3 = c$, $W_4 = s$.

Step 1: elementary partials.

$$\nabla D = (2x_1, 2x_2, 0, 0), \quad \nabla N_c = (-x_3, x_4, -x_1, x_2), \quad \nabla N_s = (-x_4, -x_3, -x_2, -x_1). \quad (27)$$

Step 2: gradients of θ_c, θ_s (quotient rule). From $\theta_c = N_c/D$,

$$\frac{\partial \theta_c}{\partial x_i} = \frac{(\partial_i N_c) D - N_c (\partial_i D)}{D^2} = \frac{\partial_i N_c}{D} - \theta_c \frac{\partial_i D}{D}, \quad (28)$$

using $N_c/D^2 = \theta_c/D$, and likewise for θ_s . Component-wise,

$$\nabla \theta_c = \frac{1}{D} \begin{pmatrix} -x_3 - 2x_1 \theta_c \\ x_4 - 2x_2 \theta_c \\ -x_1 \\ x_2 \end{pmatrix}, \quad \nabla \theta_s = \frac{1}{D} \begin{pmatrix} -x_4 - 2x_1 \theta_s \\ -x_3 - 2x_2 \theta_s \\ -x_2 \\ -x_1 \end{pmatrix}, \quad (29)$$

the x_3, x_4 columns lacking a ∂D term because $\partial_3 D = \partial_4 D = 0$.

Step 3: the Jacobian $J_{ji} = \partial W_j / \partial x_i$. Rows $j = 3, 4$ vanish (W_3, W_4 are constant in $\mathbf{x}$). For rows 1, 2, since c, s are constants,

$$\frac{\partial W_1}{\partial x_i} = c \frac{\partial \theta_c}{\partial x_i} + s \frac{\partial \theta_s}{\partial x_i}, \quad \frac{\partial W_2}{\partial x_i} = c \frac{\partial \theta_s}{\partial x_i} - s \frac{\partial \theta_c}{\partial x_i}. \quad (30)$$

Explicitly (every entry carries a factor $1/D$),

$$\begin{aligned} \partial_1 W_1 &= \frac{1}{D} [c(-x_3 - 2x_1 \theta_c) + s(-x_4 - 2x_1 \theta_s)], & \partial_2 W_1 &= \frac{1}{D} [c(x_4 - 2x_2 \theta_c) + s(-x_3 - 2x_2 \theta_s)], \\ \partial_3 W_1 &= \frac{1}{D} [-(cx_1 + sx_2)], & \partial_4 W_1 &= \frac{1}{D} [cx_2 - sx_1], \end{aligned} \quad (31)$$

$$\begin{aligned} \partial_1 W_2 &= \frac{1}{D} [c(-x_4 - 2x_1 \theta_s) - s(-x_3 - 2x_1 \theta_c)], & \partial_2 W_2 &= \frac{1}{D} [c(-x_3 - 2x_2 \theta_s) - s(x_4 - 2x_2 \theta_c)], \\ \partial_3 W_2 &= \frac{1}{D} [sx_1 - cx_2], & \partial_4 W_2 &= \frac{1}{D} [-(cx_1 + sx_2)]. \end{aligned} \quad (32)$$

Step 4: the chain-rule term $J_{\mathbf{W}}^\top \mathbf{x}$. Only $j = 1, 2$ contribute to $(J_{\mathbf{W}}^\top \mathbf{x})_i = \sum_j x_j \partial_i W_j$:

$$\begin{aligned} (J_{\mathbf{W}}^\top \mathbf{x})_i &= x_1 (c \partial_i \theta_c + s \partial_i \theta_s) + x_2 (c \partial_i \theta_s - s \partial_i \theta_c) \\ &= \underbrace{(cx_1 - sx_2)}_A \partial_i \theta_c + \underbrace{(sx_1 + cx_2)}_B \partial_i \theta_s. \end{aligned} \quad (33)$$

Substituting the columns of $\nabla \theta_c, \nabla \theta_s$,

$$J_{\mathbf{W}}^\top \mathbf{x} = \frac{1}{D} \begin{pmatrix} A(-x_3 - 2x_1 \theta_c) + B(-x_4 - 2x_1 \theta_s) \\ A(x_4 - 2x_2 \theta_c) + B(-x_3 - 2x_2 \theta_s) \\ -(Ax_1 + Bx_2) \\ Ax_2 - Bx_1 \end{pmatrix}. \quad (34)$$

Step 5: simplifying components 3 and 4.

$$Ax_1 + Bx_2 = (cx_1 - sx_2)x_1 + (sx_1 + cx_2)x_2 = cx_1^2 - sx_1x_2 + sx_1x_2 + cx_2^2 = c(x_1^2 + x_2^2), \quad (35)$$

$$Ax_2 - Bx_1 = (cx_1 - sx_2)x_2 - (sx_1 + cx_2)x_1 = cx_1x_2 - sx_2^2 - sx_1^2 - cx_1x_2 = -s(x_1^2 + x_2^2). \quad (36)$$

Hence, with $\mathbf{g} = \mathbf{W} + J_{\mathbf{W}}^\top \mathbf{x}$ and $D - (x_1^2 + x_2^2) = \epsilon$,

$$g_3 = c - \frac{c(x_1^2 + x_2^2)}{D} = c \frac{\epsilon}{D}, \quad g_4 = s \frac{\epsilon}{D}, \quad (37)$$

recovering the boxed identity (17). Finally, since every entry above is of the form (phase-independent) $\cdot c + (\text{phase-independent}) \cdot s$, one has $\mathbf{g} = c \mathbf{v}_1 + s \mathbf{v}_2$ as in (14), giving the rank-2 information matrix of Sec. 3.1.